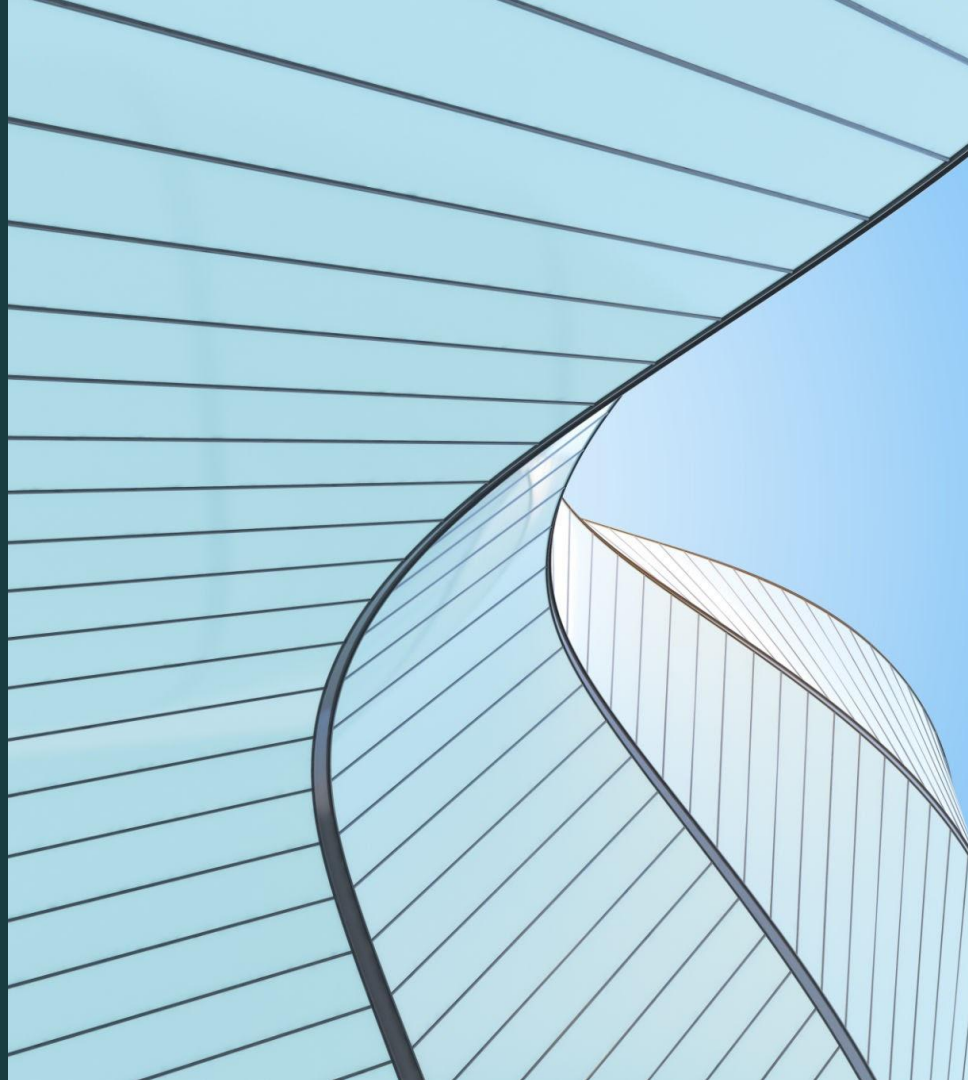


Simple Stopwatch App

CS 3020 OOP in c#
Noelle Saldana



Full code

Form1.cs

TimerApp

```
1 using System.Diagnostics;
2
3 namespace TimerApp {
4
5     //Interface for stopwatch, behaviors: Start, Stop, Reset, state variables: Time
6     public interface IStopwatchService {
7         void Start();
8         void Stop();
9         void Reset();
10        TimeSpan GetTimeElapsed();
11        bool IsRunning { get; }
12    }
13
14    //Base class for stopwatch, implements IStopwatchService, uses System.Diagnostics
15    public abstract class StopwatchBase : IStopwatchService {
16        protected System.Diagnostics.Stopwatch stopwatch;
17
18        public bool IsRunning => stopwatch.IsRunning;
19
20        public StopwatchBase() {
21            stopwatch = new System.Diagnostics.Stopwatch();
22        }
23
24        public virtual void Start() {
25            stopwatch.Start();
26        }
27
28        public virtual void Stop() {
29            stopwatch.Stop();
30        }
31
32        public virtual void Reset() {
33            stopwatch.Reset();
34        }
35
36        public virtual TimeSpan GetTimeElapsed() {
37            return stopwatch.Elapsed;
38        }
39    }
40
41    public class ActualStopwatch : StopwatchBase {
42        //Empty, all behaviors in base class
43    }
44
```

<- Interface

<- Abstract class handles behavior

<- Concrete Class

Form1.cs

TimerApp

StopwatchBase

```
44
45 public partial class Form1 : Form {
46
47     private IStopwatchService stopwatch;
48
49     //Load the window and create the stopwatch
50     public Form1() {
51         InitializeComponent();
52         stopwatch = new ActualStopwatch();
53     }
54
55     //Update the time passed and the winforms label
56     private void timer1_Tick(object sender, EventArgs e) {
57         TimeSpan ts = stopwatch.GetTimeElapsed();
58         labelTime.Text = string.Format("{0:00}:{1:00}:{2:00}.{3:000}", ts.Hours, ts.Minutes, ts.Seconds, ts.Milliseconds);
59     }
60
61     //Starts the stopwatch
62     private void Start_Button_Click(object sender, EventArgs e) {
63         stopwatch.Start();
64         timer1.Start();
65     }
66
67     //Pauses the stopwatch
68     private void Stop_Button_Click(object sender, EventArgs e) {
69         stopwatch.Stop();
70         timer1.Stop();
71     }
72
73     //Resets the time and stops the stopwatch
74     private void Reset_Button_Click(object sender, EventArgs e) {
75         stopwatch.Reset();
76         labelTime.Text = "00:00:00.00";
77     }
78
79     private void labelTime_Click(object sender, EventArgs e) {
80         //Empty because I accidentally double clicked the label in the designer,
81         //thus creating this event handler,
82         //and the program wont run without it... oops
83     }
84
85
```

<- App window

<- Events

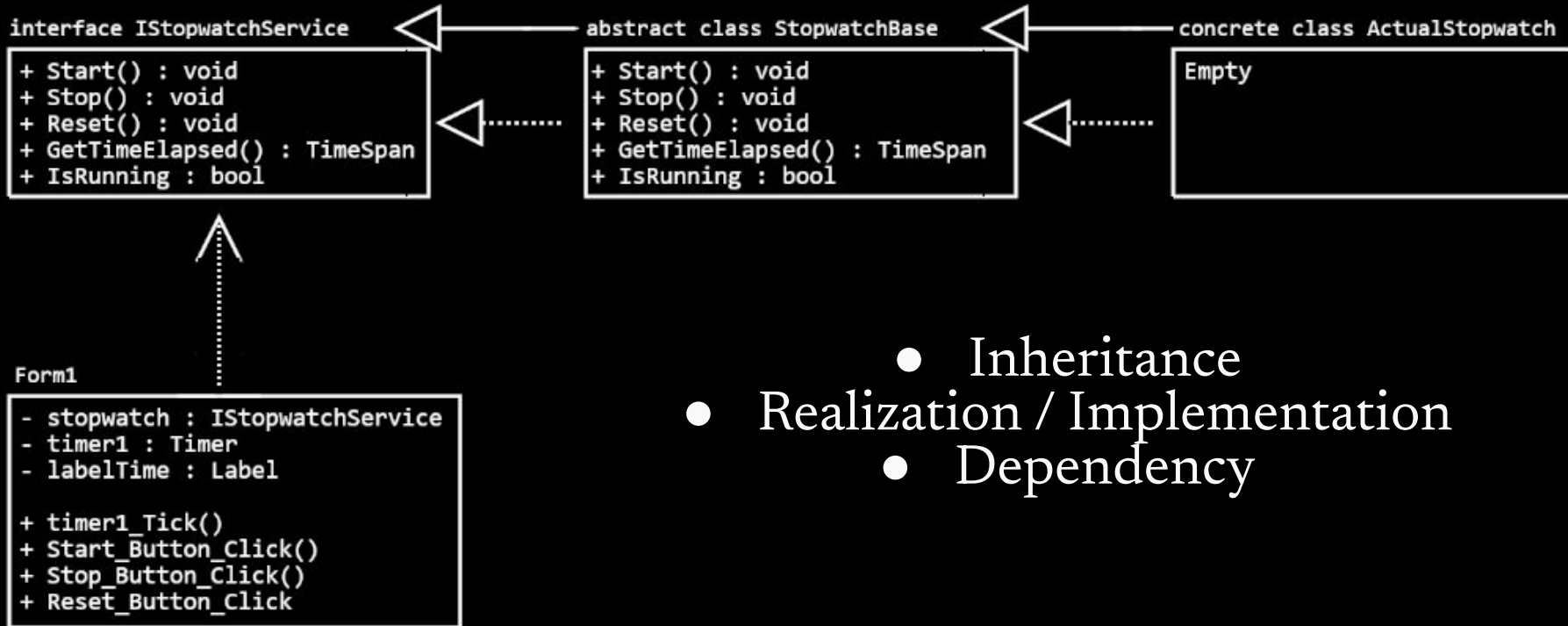
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UML Diagram ...plus some minor Winforms grievances



- Inheritance
- Realization / Implementation
- Dependency